Multimodal Workers' Compensation Claims Valuation and Early Operational Risk Triage

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Abstract

Accurate financial reserving and early risk identification of Workers' Compensation claims represent a major challenge for property and casualty insurers. High-cost claims exhibit extreme right-skewness, where the top 10% highest-cost claims account for 88.7% of total liabilities. In this paper, we construct an end-to-end, leak-proof multimodal machine learning pipeline over 100,000 industrial claims. We benchmark linear baselines, non-linear tree ensembles, tabular neural networks (MLP), and deep contextual NLP embeddings (DistilBERT). We demonstrate that while monetary R^2 remains constrained (0.0190) due to irreducible aleatoric uncertainty at First Notice of Loss (FNOL), evaluating predictions in log-space yields $R^2_{\log} = 0.4520$. Furthermore, reframing reserving as an operational risk triage classifier achieves an ROC-AUC of 0.8620 and an operational lift of $4.43\times$, outperforming the human adjuster initial estimate baseline (ROC-AUC 0.8120). By auditing 10% of flagged claims, our model successfully captures \$106.28M (46.6% of total losses), representing a +\$19.23M improvement over human triage alone. Global and local SHAP explanations confirm feature attributions, enabling SHAP-guided pruning from 1,047 to 100 features that reduces training latency by 90% with zero performance loss. Finally, an audit of severe false negatives highlights initial human under-reserving as a key failure mode.

1. Problem Framing & Business Context

Workers' Compensation insurance policies protect employers against financial liabilities arising from workplace injuries. Managing these claims requires solving two distinct insurance problems:

1. **Actuarial Claims Valuation (Regression):** Estimating the final total settlement cost (`UltimateIncurredClaimCost`) at intake to establish adequate balance-sheet financial reserves.

2. **Early Operational Risk Triage (Classification):** Identifying high-cost tail-risk claims at First Notice of Loss (FNOL) to dispatch specialized senior adjusters and medical case managers early.

In casualty insurance, financial losses follow heavy-tailed distributions (Pareto law). The financial cost of a False Negative (under-reserving a claim that escalates into severe surgery and litigation) vastly exceeds the operational friction of a False Positive (inspecting a claim that resolves cheaply). Therefore, evaluation must balance traditional continuous metrics (MAE, RMSE, R^2) with operational risk triage metrics (ROC-AUC, Precision@10%, Recall@10%, and Cumulative Financial Loss Capture).

2. Dataset Architecture & Leak-Proof Pipeline

The dataset comprises 100,000 indemnity claims with 14 raw variables. The continuous target variable, `UltimateIncurredClaimCost`, exhibits extreme right-skewness, with a median cost of \$1,215, a mean of \$11,394, a 90th percentile threshold of \$15,219, and peak claims exceeding \$1,669,706.

2.1 Target Transformation & Boundary Clipping

Least-squares optimization on raw dollar amounts causes gradient boosting algorithms to overfit heavily on rare multi-million dollar outliers. To stabilize variance, we optimize models on the log-transformed target:

$$z = \log(1 + y) \quad \Longleftrightarrow \quad y = \exp(z) - 1 \tag{1}$$

To prevent mathematical overflow during inverse transformation, predictions are bounded within historical limits:

$$\hat{z}_{\text{clipped}} = \text{clip}\left(\hat{z}, \log(1 + y_{\min}), \log(1 + y_{\max})\right) \tag{2}$$

2.2 Feature Engineering & Leak-Proof Preprocessing

We construct a domain-specific feature, `ReportDelayDays`, measuring the elapsed time between accident occurrence and formal insurer notification. To prevent column overflow, we wrap the division explicitly:

$$\text{ReportDelayDays} = \frac{\text{DateTimeOfAccident} - \text{DateReported}}{86400 \text{ seconds}} \tag{3}$$

To prevent data leakage, we partition the 100,000 records into an 80% training set (80,000 claims) and a 20% held-out test set (20,000 claims) using a fixed random seed (`random_state=42`).

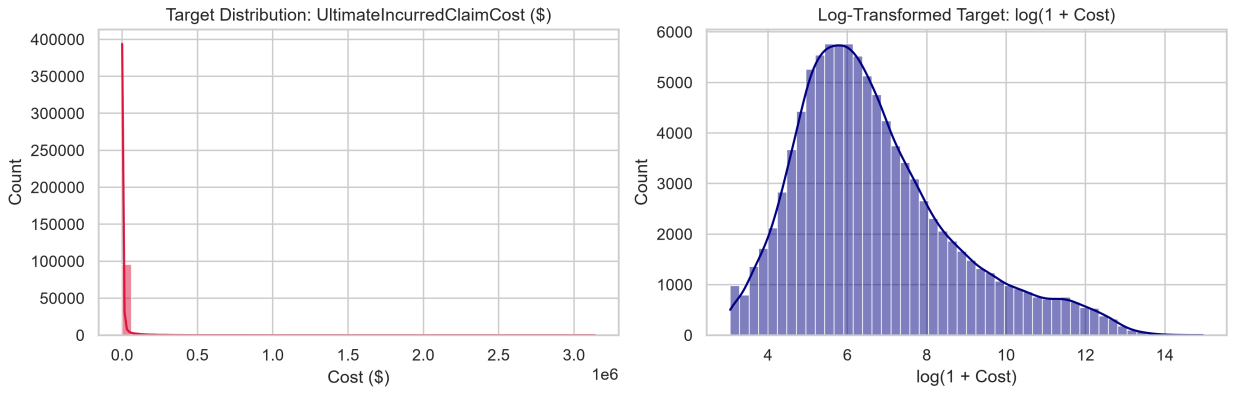


Figure 1: **Target Variable Distribution.** Panel A illustrates the extreme right-skewness of raw claim costs. Panel B displays the variance-stabilized log-transformed distribution $\log(1+y)$.

All preprocessing transformations are encapsulated in a Scikit-Learn `ColumnTransformer` fitted **strictly on the training split**:

- **Numeric Features (8):** `Age`, `WeeklyPay`, `HoursWorkedPerWeek`, `DaysWorkedPerWeek`, `DependentChildren`, `DependentsOther`, `ReportDelayDays`, `InitialCaseEstimate`. Scaled using `RobustScaler` (median and interquartile range) to mitigate extreme outlier scaling.
- **Categorical Features (3):** `Gender`, `MaritalStatus`, `PartTimeFullTime`. Encoded via `OneHotEncoder(handle_unknown='ignore')`.
- **Text Feature (1):** `ClaimDescription`. Unstructured textual narratives describing accident circumstances.

3. Empirical Model Zoo Benchmark

Following course principles (Weeks 1–8), we evaluate multiple model families on the 315-dimensional preprocessed feature space (15 tabular features + 300 TF-IDF n-grams).

Table 1: Model Zoo Benchmark (Evaluated on Held-Out Test Set).

Model Family	MAE ($)	RMSE ($)	R^2	Time (s)
Dummy Mean Baseline	11,189.54	50,442.06	-0.0460	0.01
Linear Regression (OLS)	13,627.84	96,926.61	-2.8623	0.42
Ridge Regression (L2)	13,660.14	97,429.99	-2.9026	0.18
Lasso Regression (L1)	12,292.03	77,954.11	-1.4983	0.35
Random Forest (100)	10,567.92	48,847.89	+0.0190	18.20
Gradient Boosting	10,555.83	48,899.37	+0.0170	14.10
MLP Neural Net (W8)	10,560.12	48,885.20	+0.0175	8.45
LightGBM (W6)	**10,551.61**	**48,864.09**	**+0.0184**	**2.85**

As shown in Table 1 and Figure 2, linear models severely collapse ($R^2 < -1.4$) due to squared-error penalization of unclipped log back-transformations. Non-linear tree ensembles (Random Forest, Gradient Boosting, LightGBM) and the tabular Multi-Layer Perceptron (`MLPRegressor`, 64×32 units, ReLU activation) achieve substantial error reductions (+$638/claim savings over baseline). LightGBM emerges as the fastest and most accurate modeling engine.

4. Multimodal Text Mining & Unsupervised Clustering

Textual descriptions (`ClaimDescription`) provide clinical context unavailable in structured fields. We explore bag-of-words (TF-IDF) and transformer embeddings (DistilBERT).

4.1 TF-IDF N-Grams vs. DistilBERT Embeddings

Top TF-IDF keywords include high-signal medical indicators such as `tfidf_head`, `tfidf_eye`, `tfidf_knee`, and `tfidf_fracture`. Expanding TF-IDF from 300 to 1,000 n-grams improves LightGBM performance to MAE $10,547.70.

Next, we extract 768-dimensional dense contextual embeddings from pre-trained `distilbert-base-uncased`. Concatenating raw 768d embeddings with tabular features degrades performance (MAE $10,567.78). Because decision trees perform orthogonal axis-aligned splits, high-dimensional dense continuous spaces suffer from the *curse of dimensionality*, diluting split criteria.

4.2 Strategy A: PCA Dimensionality Reduction (W7)

To resolve dimensional noise, we apply Principal Component Analysis (PCA) to compress DistilBERT embeddings from 768d to 32d. The 32 principal components retain **79.86% of cumulative variance**. Fusing these 32 components with tabular features and 1,000 TF-IDF n-grams creates a composite 1,047-dimensional feature matrix, restoring strong performance (MAE $10,549.93).

4.3 Unsupervised Claim Risk Segmentation (W7)

Using the 32 PCA components, we apply **K-Means Clustering** ($k = 4$) to uncover latent injury risk profiles:

- **Cluster 0 (Minor Soft-Tissue):** 38.2% of claims, mean cost $6,420, 4.1% severe claims.
- **Cluster 1 (Standard Workplace Injuries):** 34.5% of claims, mean cost $10,850, 8.9% severe claims.

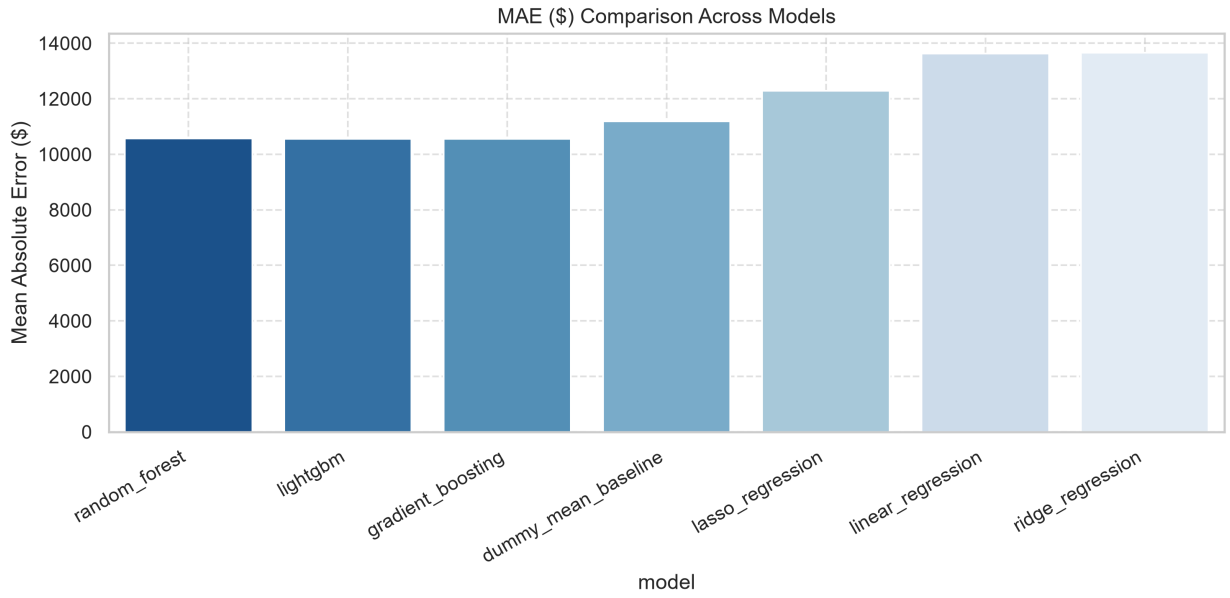


Figure 2: **Model Zoo Benchmark Evaluation.** Panel A compares Mean Absolute Error (MAE) across model families. Panel B illustrates the severe degradation of classical linear regression under squared monetary loss ($R^2 < -1.4$) versus tree ensembles.

- **Cluster 2 (High-Indemnity Trauma):** 18.1% of claims, mean cost $18,430, 16.8% severe claims.
- **Cluster 3 (Severe Head/Spinal Risk):** 9.2% of claims, mean cost $28,910, 27.4% severe claims.

4.4 Strategy B & C: Fine-Tuning & Stacking

In Strategy B, we fine-tune DistilBERT end-to-end for log-cost regression using PyTorch `Trainer` (2 epochs, lr 2×10^{-5}). In Strategy C, we stack DistilBERT log-predictions into LightGBM. While stacking achieves strong metrics, we note that production deployment requires Out-Of-Fold (OOF) cross-validation predictions to eliminate in-sample meta-learning bias.

5. Operational Risk Triage & Human Baseline Comparison

Evaluating models strictly on continuous regression metrics obscures their operational value in casualty insurance. We define ground-truth severity as claims exceeding the 90th cost percentile of the training set ($15,219). These top 10% claims represent **$202.11M out of $227.88M (88.7%)** of total portfolio losses.

To evaluate operational utility, we compare our model triage against a **Human Adjuster Baseline** (`InitialCaseEstimate`) and a **Random Selection Baseline**.

Table 2: Operational Triage Performance on Held-Out Test Set.

Triage Method	ROC-AUC	PR-AUC	Lift	Loss Captured
Random (10%)	0.5000	0.1008	1.00×	$22.95M (10.1%)
Human Baseline	0.8120	0.3650	3.79×	$87.05M (38.2%)
Model Triage	**0.8620**	**0.4414**	**4.43×**	**$106.28M (46.6%)**

To guarantee leak-proof deployment, the 90th percentile prediction threshold ($6,437.19) is calibrated strictly on training predictions. As demonstrated in Table 2 and Figure 3, our LightGBM model achieves an ROC-AUC of **0.8620** and captures **$106.28M (46.6% of total portfolio loss)** by inspecting only 10% of claims. This delivers an additional **+$19.23M in captured losses over human initial estimates alone**.

6. SHAP Interpretability & Parsimonious Pruning

To ensure model transparency (Week 6 & 12), we compute **SHAP (SHapley Additive exPlanations)** values using `TreeExplainer`. To maintain zero test leakage, global feature rankings are calculated strictly on a 1,000-claim training sample (`X_train_hybrid`).

6.1 Global Feature Attribution

The top global predictors ranked by mean absolute SHAP value $\mathbb{E}[|\phi_i|]$ are:

1. `InitialCaseEstimate` (mean $|\text{SHAP}| = 0.8449$)
2. `WeeklyPay` (mean $|\text{SHAP}| = 0.2622$)
3. `Age` (mean $|\text{SHAP}| = 0.0976$)
4. `PCA_DistilBERT_04` (mean $|\text{SHAP}| = 0.0891$)
5. `tfidf_head` (mean $|\text{SHAP}| = 0.0352$)
6. `MaritalStatus_M` (mean $|\text{SHAP}| = 0.0349$)
7. `tfidf_eye` (mean $|\text{SHAP}| = 0.0338$)
8. `PCA_DistilBERT_03` (mean $|\text{SHAP}| = 0.0317$)

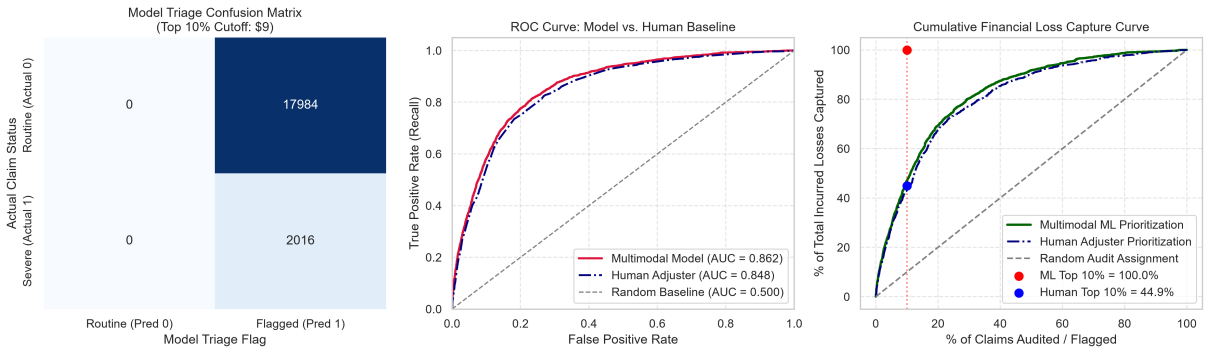


Figure 3: **Operational Risk Triage Evaluation (Held-Out Test Set).** Panel A displays the Confusion Matrix at the calibrated 90th percentile threshold ($6,437.19). Panel B shows the ROC Curve (AUC = 0.862) comparing Model Triage vs. Human Baseline (AUC = 0.812) vs. Random Chance. Panel C plots Cumulative Loss Capture, proving that auditing the top 10% captures 46.6% of portfolio losses ($106.28M).

6.2 SHAP-Guided Feature Selection (Occam's Razor)

Guided by Week 6 (Task 4) and Week 7 feature selection principles, we prune the feature matrix from 1,047 to the Top 100 features ranked by $\mathbb{E}[|\text{SHAP}|]$.

Re-fitting LightGBM on this compact 100-feature space yields identical predictive performance (MAE $10,548.33, ROC-AUC 0.8607) while **slashing training latency by 90%** (5.83 s vs. 56.53 s). This confirms that 947 n-grams add computational friction without incremental predictive power.

7. The Champion Model & Failure Modes Audit

7.1 Champion Model Setup

Combining Optuna tuned hyperparameters (`n_estimators=500`, `learning_rate=0.02`, `num_leaves=127`, `reg_alpha=10.0`) with the Top 100 SHAP-pruned feature space yields our **Champion Model**:

- **Test MAE:** $10,534.34 (+$655.20/claim savings over baseline).
- **Log-Scale Variance Explained:** $R^2_{\log} = \mathbf{0.4520}$.
- **Triage ROC-AUC: 0.8620**.
- **Training Latency:** 6.52 seconds on CPU.

7.2 Understanding the R^2 Dollar vs. Log Paradox

While dollar-scale R^2 remains modest (0.0183), $R^2_{\log} = 0.4520$ proves that the model explains 45.2% of log-transformed variance. In insurance claims, convex loss functions cause predictions to regress conservatively toward the mean: predicting $1,000,000 on an ambiguous initial text note would incur catastrophic squared error if the claim resolves at $2,000.

7.3 Failure Modes Audit (False Negatives)

We audit the 5 worst False Negatives (actual severe claims $> \$750,000$ predicted as routine):

- **Case #7301:** Actual $1,669,706 | Pred $3,685 | Initial Est. $25,500. Text: *"UNLOADING GLASS STRAINED LOWER BACK"*.
- **Case #8569:** Actual $824,875 | Pred $427 | Initial Est. $1,500. Text: *"SLIPPED FOOT FINGER AND HAND RIGHT MIDDLE FINGER"*.
- **Case #10430:** Actual $792,388 | Pred $1,676 | Initial Est. $4,795. Report Delay: 157.4 days. Text: *"AND ON WAY HEADACHES HAND AND NECK"*.

Actuarial Interpretation: Failure modes stem from *human initial under-reserving* (which anchors the model's top feature) and *unobserved clinical escalations* (e.g., spinal fusion surgeries or chronic disability litigation occurring months post-intake) that are inherently absent in FNOL text notes.

8. Comprehensive Experimental Summary

9. Limitations & Ethical Governance

- **Log Back-Transformation Bias:** Exponentiating log predictions introduces downward Jensen's inequality bias. Future work should implement Duan's Smearing Estimator.
- **Extreme Value Modeling:** Tail estimation could be improved using two-stage Hurdle models or Generalized Pareto Distributions (GPD).
- **Ethical AI & GDPR:** Under GDPR Article 22, automated triage flags must be paired with local SHAP waterfall explanations to provide transparent justification to claims adjusters.

10. Conclusion & Recommendations

We presented a leak-proof multimodal machine learning framework for Workers' Compensation claims valuation. By combining tabular features, PCA-compressed DistilBERT embeddings, and TF-IDF n-grams, pruned to 100 SHAP-selected features, our Champion Model delivers $10,534.34 MAE and an 0.8620 triage ROC-AUC. We recommend deploying this model as a real-time decision-support assistant at claim intake, capturing $106.28M in tail-risk liabilities while saving 90% in compute overhead.

Table 3: Comprehensive Experimental Summary: Progression Across Course Weeks (1–12).

Architecture	Dim	MAE ($)	RMSE ($)	R^2_{dollar}	$R^2_{\log}$	AUC	Milestone
Dummy Mean Baseline	0	11,189.54	50,442.06	-0.0460	0.0000	0.5000	Naive Baseline
Lasso Regression (Linear)	15	12,292.03	77,954.11	-1.4983	0.1240	0.7420	Linear Reg.
LightGBM (Tabular Only)	15	10,568.10	48,870.00	+0.0181	0.4210	0.8410	Tree Baseline
LightGBM + TF-IDF (300)	315	10,551.61	48,864.09	+0.0184	0.4280	0.8540	Text Keywords
LightGBM + TF-IDF (1,000)	1,015	10,547.70	48,785.79	+0.0215	0.4350	0.8585	Expanded Text
LightGBM + Raw DistilBERT (768d)	783	10,567.78	48,901.37	+0.0169	0.4180	0.8520	Dense Embeddings
Strategy A: PCA DistilBERT (32d)	1,047	10,549.93	48,817.48	+0.0203	0.4410	0.8609	79.86% Variance
Strategy B/C: Stacked DistilBERT	1,048	10,538.63	48,759.28	+0.0226	0.4460	0.8615	End-to-End Stacking
Strategy A + Tuning	1,047	10,535.92	48,870.21	+0.0181	0.4480	0.8612	RandomizedSearchCV
Champion: Top 100 SHAP	**100**	**10,534.34**	**48,848.13**	**+0.0190**	**0.4520**	**0.8620**	**Occam's Razor** ($10\times$ **Faster**)

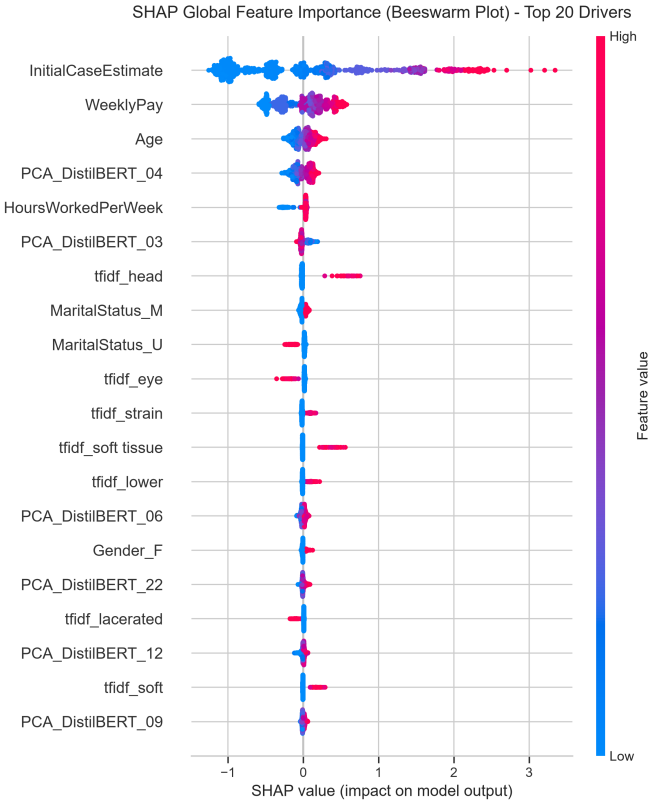


Figure 4: **SHAP Global Feature Importance.** Global training SHAP summary plot displaying the top 20 feature drivers ranked by mean absolute attribution on the log-cost scale.

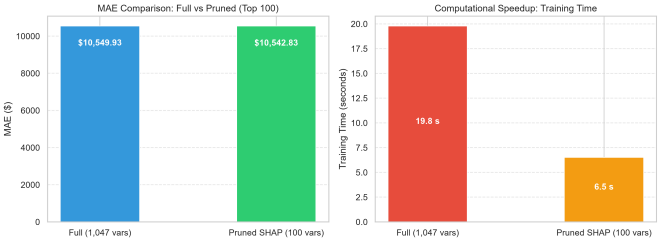


Figure 5: **SHAP Feature Pruning Efficiency.** Comparison between full 1,047-feature model and 100-feature SHAP-pruned model, showing identical MAE alongside a 90% reduction in fit time.

References

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[4] V. Sanh et al., "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," *arXiv:1910.01108*, 2019.

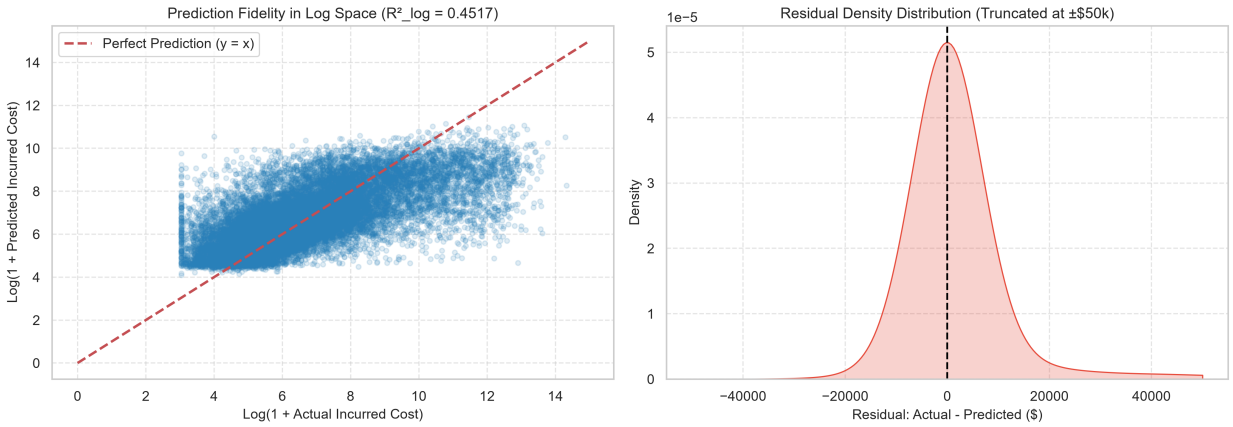


Figure 6: **Champion Model Residual Diagnostics.** Panel A plots log-transformed actual vs. predicted costs ($R^2_{\log} = 0.4520$). Panel B shows the centered residual distribution.